

DreamBooth: Fine-Tuning Text-to-Image Diffusion Models for Subject-Driven Generation

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DreamBooth (Ruiz et al., CVPR 2023) [1] is a novel method for generating personalized images of a specific subject using only 3-5 reference images, and then recreating that exact subject in new settings according to text prompts (Fig. 1). While standard text-to-image models, such as Stable Diffusion, can reliably produce generic images of a class (e.g. "a dog" or "a teapot"), they struggle to generate the same exact subject across different prompts. DreamBooth targets this gap by fine-tuning a pretrained model so that a rare identifier token, such as "sks," becomes linked with a specific subject's visual characteristics. This is necessary because even the most descriptive prompts restrict the level of detail that a visual image provides. For instance, describing a dog's fur pattern can still be interpreted in many color shades or orientations, limiting how precisely the model can reconstruct the same subject in new contexts. A key challenge is that simply fine-tuning on a few subject images can cause language drift, in that the class label "dog" starts referring only to the trained subject, wiping the diversity within the class that defines a general dog. DreamBooth addresses this by adding a prior preservation loss that trains on generic class images alongside subject images, preventing the model from overwriting its broader class knowledge.

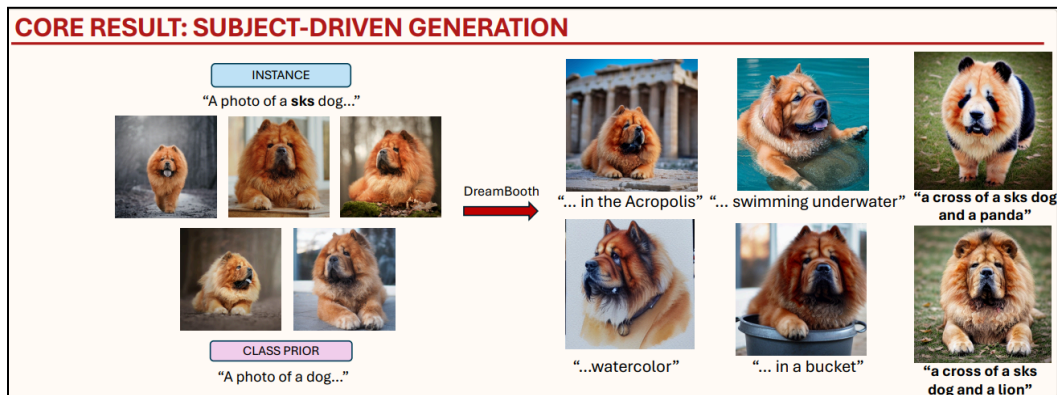


Figure 1

Our reimplementation targets three results from the paper to demonstrate DreamBooth's core mechanisms: **binding subjects to unique tokens** and **class-prior preservation**. First, we reproduced the quantitative metrics from Table 1, using DINO and CLIP-I for subject fidelity and CLIP-T for prompt fidelity. We then tested the prior preservation ablation from Table 3 and Figure 6, which directly tests whether the prior loss prevents class collapse and its sensitivity. Finally, we generated the learned subject in new scenes or artistic styles to visually confirm the subject's identity was preserved. We also went beyond the scope of the paper, including by trying to break assumptions of the method to observe its response: giving one token multiple subject identities, merging two species into one subject via text prompts, and applying DreamBooth to a human subject (consent granted by Professor Weinberger).

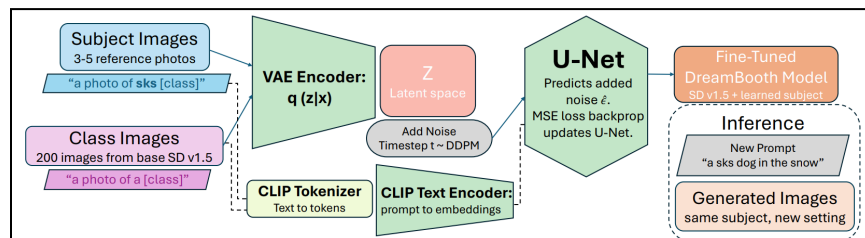


Figure 2

The architecture (Fig. 2) was built around Stable Diffusion [2] v1.5's individual components. The original paper used Google's proprietary Imagen model for their primary results but also reported Stable Diffusion numbers, so this was our comparison baseline. The model consists of a VAE for converting images into latent space, a CLIP tokenizer and text encoder for converting prompts into conditioning embeddings, a U-Net denoiser for predicting noise, and a DDPM noise scheduler. In our main experiments, the VAE and CLIP text encoder were frozen, and only the U-Net was updated. Freezing the VAE kept the latent representation stable, since its learned image-to-latent mappings were already reliable. Similarly, freezing the text encoder reduced the risk of overfitting considering how few training images we had. Therefore, the U-Net was the only component being updated according to the gradients during training. It learned the proper mapping of the rare token embedding for 'sks' to the specific subject's visual features, so that prompts containing 'sks dog' denoised toward that exact dog rather than a generic one.

At each training step, images were first resized and cropped to 512*512, encoded into latent space by the frozen VAE, and noise was added at a randomly sampled timestep. The U-Net then received the noisy latent, the timestep, and a CLIP text embedding from either the instance prompt ("a photo of sks [class]") or the class prompt ("a photo of a [class]"), and predicted the noise that was added. Therefore, the training objective was minimizing the mean squared error between the predicted and actual noise.

We tested the effect of **prior preservation** (Fig. 3) by running the model both while including and excluding it. With prior preservation enabled, both instance and class images were concatenated into a single batch and processed in one U-Net forward pass. The predictions were then split to compute separate losses: $L_{total} = L_{instance} + \lambda * L_{prior}$, with $\lambda = 1.0$ (matching the paper's default). From the official Google DreamBooth dataset [3], four subjects that span the paper's visual categories: dog (corgi), bear_plushie, colorful_sneaker, and teapot were used (with 4-6 reference images from each). For prior preservation, we generated 200 class-prior images per class noun from the frozen base model using text prompts like "a photo of a dog."

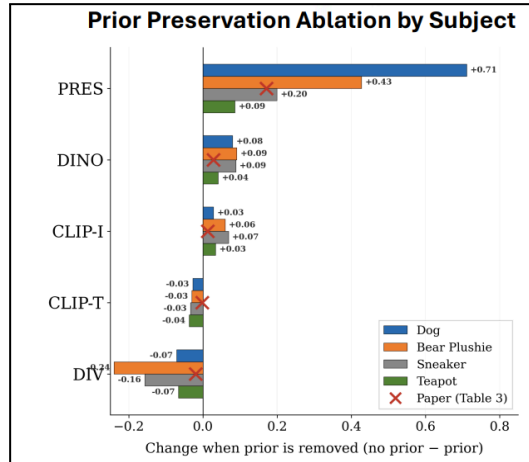


Figure 3

For each subject, we trained 8 runs with prior preservation either being enabled (on) or disabled (off). For the dog, we added three ablations: $n = 3$ and $n = 4$ training images, and unfreezing the text encoder at a lower learning rate of $2e-6$ (for 11 total runs). We ran experiments at two U-Net learning rates: $5e-6$, matching the paper's reported setting, and $3e-6$ as another alternative (suggested by the paper). All runs used batch size 1, 800 steps, AdamW optimizer, and seed 42 to ensure results were comparable. We scored five metrics: DINO and CLIP-I for subject fidelity, CLIP-T for prompt fidelity, PRES for class-prior collapse (lower is better), and DIV for output diversity (higher is better).

Our results (Table 1, right) verified the 5e-6 run closely reproduced the paper's Stable Diffusion metrics with subject fidelity being slightly below the paper (DINO 0.654 vs. 0.668, CLIP-I 0.787 vs. 0.803), while prompt fidelity was slightly higher (CLIP-T 0.325 vs. 0.305). The 3e-6 run shifted these values

Result	DINO	CLIP-I	CLIP-T	Interpretation
Dreambooth Paper, Stable Diffusion	0.668	0.803	0.305	Paper baseline
Ours, LR = 5e-6	0.654	0.787	0.325	Best Overall Reproduction
Ours, LR = 3e-6	0.608	0.764	0.335	Higher prompt fidelity, lower subject fidelity



with DINO decreasing to 0.608 and CLIP-T increasing to 0.335, which suggests a tradeoff with the learning rate. A higher learning rate learns the subject more aggressively but risks overfitting, while a lower one preserves prompt flexibility better but sacrifices the specificity of the subject. These metrics validated the reproductions of the paper's main result, further supported with the prior preservation ablation. Without the prior loss, prompting the model with "a photo of a dog" (no "sks") generated images that looked like the previous training subject instead of diverse dogs. Table 2 (left) supports this for the dog subject with PRES increasing from 0.236 to 0.874 when prior preservation was removed, meaning outputs for the generic class drastically collapsed,

losing the generic images within the class. Diversity also decreased to 0.604, however DINO actually increased without prior (0.787 vs. 0.664), but this was due to overfitting since the model kept producing the subject even when prompted with the generic class, inflating similarity scores.

The n-image sweep on the dog showed that even three images captured recognizable identity for a common subject like the corgi, which makes sense given the base model already has a strong prior for what dogs look like. The text-encoder-unfrozen ablation at LR 2e-6 produced marginally stronger subject fidelity but introduced occasional color shifts, suggesting that updating the text encoder can help bind the token more tightly but risks destabilizing the conditioning on such a small dataset. We designed three experiments beyond the paper to test what happens when certain assumptions of DreamBooth are violated. The fusion (Figure 4) experiment that merged two subjects involved training on two different dog breeds (the corgi and a white Bichon) using the same "sks" dog token. The outputs blended traits from both, confirming that one token cannot cleanly encode two contradictory subjects. In cross-species prompt engineering, prompts like "a cross of a sks dog and a panda" generated hybrids that preserved the corgi's facial features while using the panda's black and white body, showing how the learned token interacts with the model's existing knowledge. In the human-subject experiment, we trained on photos of Professor Weinberger and generated him as a doctor, astronaut, etc. His identity was recognizable but less consistent than with the dog, which suggests human personalization is more sensitive to prompt wording and the base model's person prior.

Looking back on this project, the hardest part was figuring out what to do beyond the paper that is

OTHER APPLICATIONS: DOG BREEDS



Figure 4

both feasible and meaningful. Coming up with experiments like merging two dogs under one token or training on a person required understanding the method well enough to know where it should break, and its current assumptions. Another challenge was that the paper's best results used Imagen, which we had no access to, so we ran everything on Stable Diffusion v1.5 and compared against the paper's own SD numbers instead. The prior preservation ablation was the most important result because it confirmed the mechanism still works on a different model than the one the authors primarily used. With more time, a sweep of other λ values (0.25, 0.5, 1.0, 2.0) would help isolate how strongly the prior needs to be weighted, beyond just testing it as on or off. We would also want to run the frozen versus unfrozen text encoder comparison across all subjects instead of just the dog, and evaluate the human subject experiment more rigorously with additional people and more reference images. A comparison with methods like Custom Diffusion [4] or fine-tuning with LoRA, which use fewer trainable parameters and support multiple subjects, would also be worth exploring.

AI Disclaimer: Since DreamBooth fine-tunes a pretrained model, the base Stable Diffusion v1.5 architecture was already available. We wrote the training loop, inference pipeline, and evaluation metrics around it, with help from Claude (Anthropic) for debugging, coding, and organizing outputs into comparison grids. All experiment design and analysis was our own.

References

- [1] N. Ruiz, Y. Li, V. Jampani, Y. Pritch, M. Rubinstein, K. Aberman. "DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation." CVPR 2023. arXiv:2208.12242.
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- [3] Google DreamBooth Dataset. <https://github.com/google/dreambooth>
- [4] N. Kumari, B. Zhang, R. Zhang, E. Shechtman, J.-Y. Zhu. "Multi-Concept Customization of Text-to-Image Diffusion." CVPR 2023